FRONT VIEW



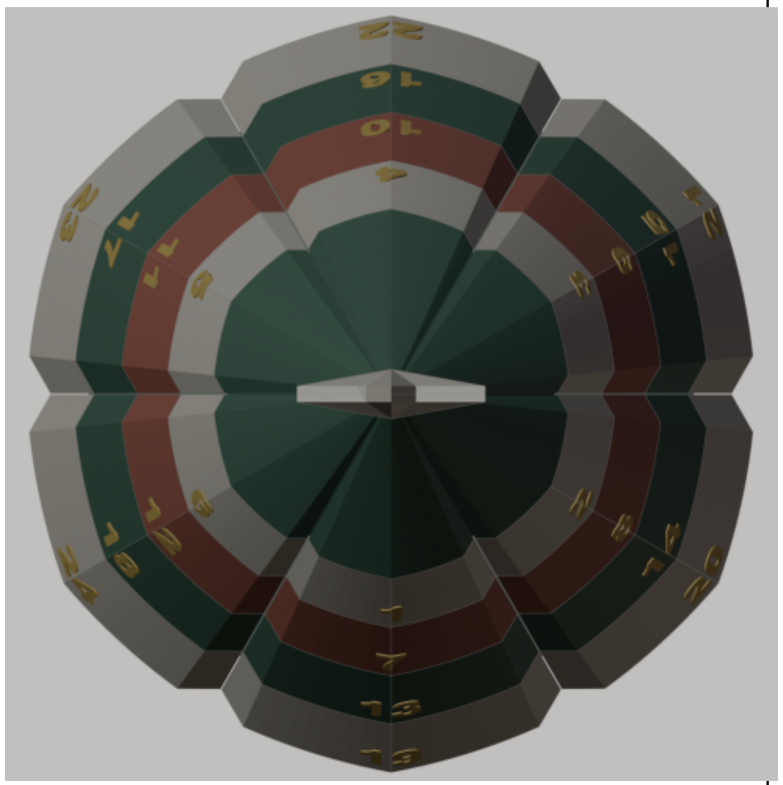
SIDE VIEW (right)



ASSEMBLED



TOP VIEW — 1 at front, numbers run left→right



Tier	Cells	Ø max	H	Overhang	Top edge Ø	Cavity
Top	secret box	134	110	15	→ star	≈400 cm³
T1	6	170	80	15	104	156 cm³
T2	6	205	80	15	140	254 cm³
T3	6	240	80	15	175	375 cm³
T4	6	275	80	15	210	520 cm³

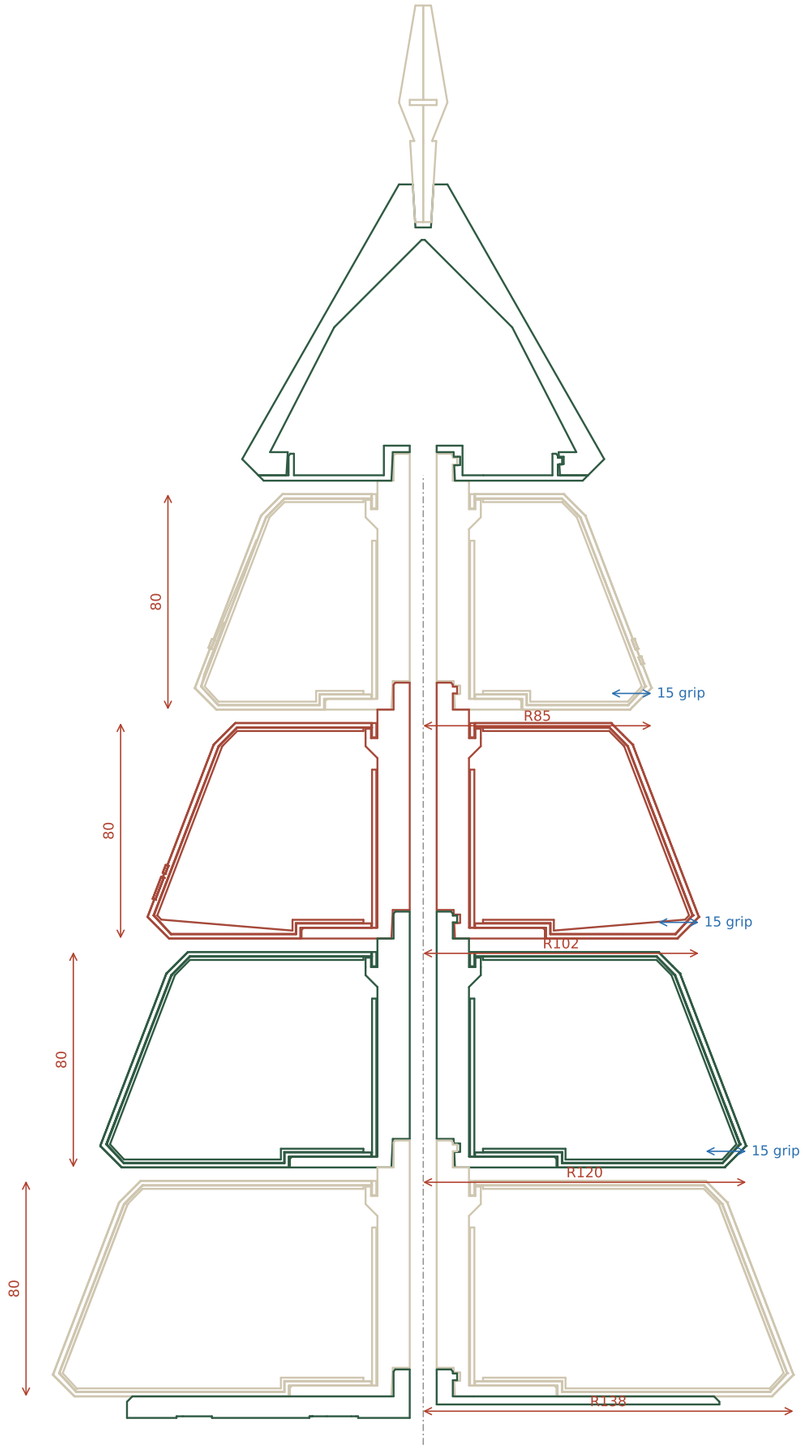
Overhang = how far a tier's widest edge sticks out past the top edge below (finger grip).
Each cell: four front facets, sharp ridge + 2 front edges, sharp sides, 21° lean, 8 mm chamfers top + bottom
Numbering: compartment 1 faces front (−Y); each tier starts at front, runs left→right
(counter-clockwise seen from above). Flip P['CLOCKWISE'] to reverse.

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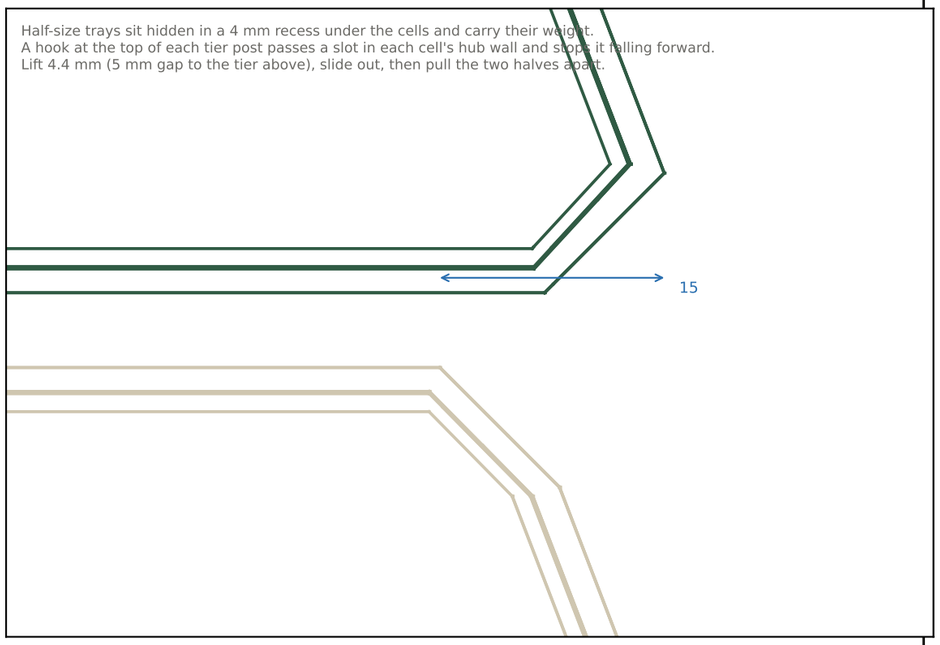
Assembly drawing — front, side, top views

Units: mm · Scale as noted · Sheet 1/3 · 2026-10-16

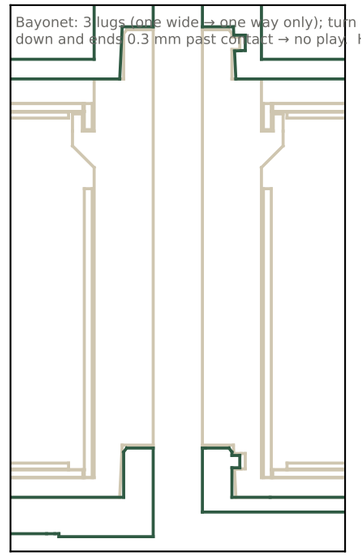
Colour = filament. Every closed outline is a separate printed part.



DETAIL B — T4 closed top, T3 tray and the 15 mm finger overhang



DETAIL C — self-tightening bayonet stack, hidden tray, trunk hook, 2:1



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Section A–A (plane X=0, through compartments 1/3, 5/7, 9/11, 13/16, 19/22)

Units: mm · Scale as noted · Sheet 2/3 · 2026-10-16

Group	Part	Qty	Filament	Print bbox X×Y×Z	Volume cm³	Orientation	STL file(s)
Core	base	1	green	220×220×18	298.6	flat on bed	Core_Base_green.stl
Core	star	2	ivory	68×80×9	8.8	flat back down	Core_Star_Back_ivory.stl
Core	top	1	green	129×134×108	182.1	cap: wide end down, open bottom	Core_TopTier_green.stl
Core	topfloor	1	green	117×122×13	28.4	secret floor: flat plate down, rim up	Core_TopFloor_green.stl
T1	carrier	1	ivory	73×73×95	83.4	tray on bed	T1_Carrier_ivory.stl
T1	module	12	ivory	80×64×42	16.1	flat radial side wall down, open side up	T1_C01…C06_*
T2	carrier	1	terra	90×90×95	92.4	tray on bed	T2_Carrier_terra.stl
T2	module	12	terra	80×81×51	22.0	flat radial side wall down, open side up	T2_C07…C12_*
T3	carrier	1	green	99×99×95	97.6	tray on bed	T3_Carrier_green.stl
T3	module	12	green	80×98×60	28.4	flat radial side wall down, open side up	T3_C13…C18_*
T4	carrier	1	ivory	99×99×95	97.6	tray on bed	T4_Carrier_ivory.stl
T4	module	12	ivory	80×115×69	35.3	flat radial side wall down, open side up	T4_C19…C24_*

Total printed parts: 57 (24 cells × 2 halves with gold number inlays, 4 tier carriers, base, top-tier floor plate + cap, star (2 halves)). No screws, magnets or glue.

ASSEMBLY
1 Put the Base on the table (Ø220, bayonet spigot up).
2 Drop the T4 carrier onto the spigot (the wide lug only fits one slot) and turn it 40° clockwise until it locks tight.
3 Press each T4 cell's half B (lip) into half A, then lower the cell over its hook so the tray drops into its recess.
4 Repeat for T3 → T1: drop, turn 40° clockwise. Then hang each tier's cells. Compartment 1 ends up at the front.
5 Put the top-tier floor plate on the T1 spigot and turn it 40° clockwise. Lower the cap over the bayonet rim of the plate and turn it 40° ANTICLOCKWISE
(reversed bayonet: opening the cap - clockwise - only tightens the floor). Join the two star halves on a ~9.5 mm piece of 1.75 filament (centre holes) and push the stem into the tip.
SECRET BOX: turn the cap clockwise 40° and lift it - the hollow cone holds ≈400 cm³ (seam hidden 2 mm above the bottom).
GARLAND: a Ø10 channel runs through the base, every tier post and the top-tier floor into the hollow cone; the wire leaves the base
through a 10×5 mm groove on its underside to the back edge (compartment 1 faces front).
OPENING A CELL: hook a finger under its overhanging edge, lift ~4.4 mm off the hook and tray, slide it straight out, then pull the two halves apart.
Any tier also lifts off whole with its cells on it.

PRINT SETTINGS (PLA / PETG)
0.4 nozzle · 0.2 layer · cell walls/floor/roof 1.6 mm = 4 lines of 0.4 · 15 % infill · no supports. Both halves print on their flat radial side wall, open side up (no supports; AdventTree_PrintParts_Orca.3mf is pre-oriented).
Fits 220×220×250. Largest parts: base Ø220 (bed limit), T4 cell half 122×80×62, top tier Ø135×110, star half 69×80×9.

FIT TUNING (in P{} at the top of the script)
BAYO_PRELOAD 0.3 / BAYO_RAMP 0.8 — bayonet tightness (too stiff → 0.15). HOOK_CLR 0.30 / TRAY_CLR 0.40. LIP_CLR 0.15 — lip of half B inside half A.
Halves fall apart → LIP_CLR 0.08. Lip won't go in → LIP_CLR 0.25.
Print one T1 cell (both halves) + T1 carrier first as a fit test (~2 h).

NUMBERS (AMS / multi-colour)
Arial Black, ~13 mm high, low on the face (7 mm above the bottom chamfer), gold, raised 0.6 mm + 0.4 mm into the wall. It is a 2nd shell of the half's mesh:
one object per half, colour per face. Open AdventTree_Assembly_Orca.3mf (painted) or AdventTree_Assembly.3mf and map gold to an AMS slot.

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Component inventory, print orientation & assembly

Units: mm · Scale as noted · Sheet 3/3 · 2026-10-16